

Neutral Value Movement

*Foundations of Rail-Agnostic Institutional Settlement:
A Technical and Governance Framework for Cross-Chain Settlement Modernization*

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Abstract

This paper introduces and develops **Neutral Value Movement (NVM)** — a conceptual and operational framework in which the economic value of a financial instrument is deliberately decoupled from any single settlement rail, network, or ledger technology. Under an NVM posture, the identity, legal standing, and economic attributes of a financial claim are treated as properties of the instrument itself, not of the infrastructure through which it happens to be held or transferred at any given moment.

The imperative for such a framework arises from the simultaneous coexistence of legacy central securities depository infrastructure (DTCC, Euroclear, Clearstream), permissioned distributed ledger platforms (JPMorgan Kinexys, Broadridge DLR, Canton Network), emerging public chain deployments (Ethereum Layer 2 networks), and conventional payment rails (Fedwire, SWIFT). In this fragmented landscape, the settlement of a cross-rail transaction today requires bespoke, bilateral engineering — an approach that scales neither operationally nor legally.

This paper makes four principal contributions: (1) a rigorous definition of rail-agnostic settlement and its distinction from interoperability; (2) the concept of cross-chain equivalence and the Equivalence Certificate as a legal-technical construct; (3) the Canonical Digital Artifact as the foundational representational standard for multi-rail financial instruments; and (4) a Multi-Rail Governance Stack with

prescriptive layers spanning standards, protocols, law, and supervision. The paper concludes with five policy recommendations addressed to regulators, central banks, market infrastructure operators, and standards bodies, and argues that settlement modernization pursued without rail-agnosticism as its foundational principle will reproduce the fragmentation it seeks to resolve.

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Section 1 — Introduction: The Settlement Rail Problem

The global settlement infrastructure for financial instruments has never been more capable — or more fragmented. In the span of fewer than ten years, the landscape has expanded from a relatively coherent, if imperfect, set of legacy central securities depositories (CSDs) and payment rails to a heterogeneous ecology of at least four distinct categories of settlement technology, each operating under different assumptions about finality, identity, governance, and interoperability.

Traditional settlement infrastructure remains the operational backbone of global capital markets. The Depository Trust & Clearing Corporation (DTCC) alone clears and settles transactions with a gross notional value exceeding \$2.5 quadrillion annually. Euroclear and Clearstream collectively hold more than €40 trillion in assets under custody and process the settlement of trillions of euros in sovereign bond and equity transactions each year. In 2024, the US equity markets migrated to T+1 settlement — a meaningful advance — yet the mechanics remain fundamentally unchanged: central counterparty clearing, netting, and deferred net settlement over messaging rails that trace their conceptual lineage to the 1960s. These systems are reliable, deeply embedded in legal frameworks, and trusted by regulators. They are also closed, slow relative to the latency expectations of modern finance, and structurally incapable of supporting intraday liquidity optimization at scale.

Against this backdrop, permissioned distributed ledger platforms have achieved genuine production scale. JPMorgan's Kinexys platform has processed over \$1.5 trillion in cumulative transaction volume since launch, averaging \$2 billion in daily throughput as of mid-2026, with near-zero settlement failure rates.

Broadridge's Distributed Ledger Repo (DLR) platform processes \$385 billion in average daily bilateral repo volume, representing the single largest production deployment of tokenized capital markets infrastructure by transaction value. The Canton Network — a privacy-enabled blockchain for institutional finance operated by Digital Asset — connects Goldman Sachs, HSBC, Microsoft, and others for synchronized cross-entity workflows. HSBC Orion handles tokenized bond issuance and settlement for sovereign and quasi-sovereign borrowers. Each of these platforms has demonstrated that distributed ledger technology is not a theoretical proposition: it is a production-grade operational reality.

Public chain deployments add a third dimension. Ethereum Layer 2 networks — Arbitrum, Optimism, Base, and others — now host regulated financial products, including BlackRock's BUIDL tokenized money market fund, which has reached nearly \$3 billion in assets under management across eight blockchain networks. Conventional payment rails — Fedwire for large-value US dollar settlement, SWIFT's cross-border messaging network connecting over 11,000 financial institutions in more than 200 countries — continue to process the overwhelming majority of global value transfer by notional volume.

The core problem that unites these developments is this: **value today is rail-dependent**. A tokenized treasury bond held in custody on JPMorgan Kinexys cannot natively settle a repo obligation on Broadridge DLR without bespoke engineering. A BlackRock BUIDL share held on Arbitrum cannot be delivered to an Euroclear account as margin collateral without a manual, legally uncertain conversion process. A SWIFT payment instruction cannot automatically trigger the release of tokenized collateral on the Canton Network without a custom integration built by the counterparties involved. In each case, the economic substance of the transaction — the transfer of value from one party to another — is held hostage to the technical and legal particularity of the rail on which the relevant asset happens to reside.

This paper proposes a solution to this structural inefficiency through the concept of **Neutral Value Movement (NVM)**: a set of protocols, governance norms, and representational standards that allows value to flow across rails without losing fidelity, identity, or legal standing. NVM is not a product, a protocol, or a platform. It is a design posture — a set of first principles that, if adopted consistently, would enable any compliant settlement infrastructure to treat the value of a financial instrument as a property of the instrument itself rather than of the rail on which it settles.

The paper's central thesis is that settlement modernization — if pursued rail by rail, platform by platform, and bilateral agreement by bilateral agreement — will reproduce the fragmentation it seeks to resolve.

The alternative requires rail-agnosticism as a foundational principle, established at the outset through standards, law, and governance architecture. This paper develops that alternative in full.

Section 2 — Rail-Agnostic Settlement

2.1 Definition and Scope

Core Concept — Definition

Rail-Agnostic Settlement

Rail-agnostic settlement is settlement that can achieve finality on any compliant execution layer, with outcome equivalence regardless of which rail executes the transaction. Under a rail-agnostic settlement model, the legal, economic, and operational outcome of a given transaction is identical whether it is executed on a permissioned DLT platform, a public blockchain, a traditional CSD, or a hybrid system — provided all rails conform to a common set of pre-settlement, matching, and finality standards.

The emphasis on *outcome equivalence* is critical and distinguishes rail-agnostic settlement from mere interoperability. Two systems may be interoperable — capable of exchanging messages and triggering actions in each other — without producing equivalent outcomes. The legal standing of finality may differ between rails; the moment at which a transfer becomes irrevocable may be defined differently; the identity attributes associated with the transferred instrument may be represented using incompatible schemas. Interoperability solves the communication problem. Rail-agnosticism solves the equivalence problem.

Institutional adoption of tokenized finance has stalled, in material part, because of the absence of rail-agnosticism. An asset manager considering the migration of a collateral pool from a legacy CSD to a tokenized platform must contend with a fundamental question: if the asset is needed on a different platform tomorrow — for margining, for repo, for payment — can it be moved without losing its legal identity, its provenance history, or its eligibility status? Where the answer is uncertain, institutions

rationality maintain parallel positions on multiple rails, compounding fragmentation rather than resolving it.

2.2 The Settlement Abstraction Layer

The mechanism through which rail-agnostic settlement becomes operationally viable is the **settlement abstraction layer** — a logical plane, above individual rails, that normalizes the pre-settlement, matching, and finality signals produced by disparate systems into a common representation. The abstraction layer does not itself execute settlement; it orchestrates the execution, monitors finality signals, and provides a canonical record of outcomes that is independent of the executing rail.

Several real-world initiatives approximate elements of this architecture. SWIFT's tokenized asset transfer experiments — spanning three phases from 2022 to 2025 — demonstrated that SWIFT's existing messaging infrastructure could act as an orchestration layer for tokenized asset transfers across multiple blockchain networks, with SWIFT providing the common identity and messaging substrate. In its Phase 3 experiments, SWIFT connected over 30 financial institutions across multiple DLT platforms, enabling the transfer of simulated tokenized assets with real-time settlement status tracking. HQLAx, which provides collateral mobility services for European sovereign and agency securities, has processed over €100 billion in collateral transactions by enabling asset swaps across multiple custody platforms without moving the underlying securities — a form of settlement abstraction applied to collateral management.

Finality's Utility Settlement Coin (USC) represents a more radical abstraction: a wholesale payment token backed by central bank reserves, designed to serve as the settlement asset for multi-rail transactions — replacing rail-specific cash legs with a single, neutral settlement instrument. Chainlink's Cross-Chain Interoperability Protocol (CCIP) provides a messaging and token transfer layer that has processed over \$7.77 billion in cross-chain transfers as of mid-2026, offering a technical substrate for the kind of cross-rail settlement signaling that an abstraction layer requires.

2.3 Settlement Finality Norms

A settlement abstraction layer is only as coherent as the finality signals it consumes. Settlement finality — the point at which a transfer becomes legally irrevocable — is defined differently across rails, creating a fundamental impediment to outcome equivalence. Three distinctions matter most.

Probabilistic versus deterministic finality. Proof-of-work blockchains achieve probabilistic finality: the likelihood that a transaction is reversed declines as subsequent blocks accumulate but never reaches zero. Modern proof-of-stake networks — Ethereum post-Merge, most permissioned DLT platforms — achieve deterministic finality: once a block is finalized, reversion is computationally and economically prohibitive by design. For regulated settlement, only deterministic finality is acceptable, which effectively excludes legacy proof-of-work chains from institutional settlement infrastructure.

Technical versus legal finality. A transaction may be technically final on a blockchain — irreversibly recorded — while remaining legally provisional under the governing law of the counterparties. The EU's DLT Pilot Regime and the UK's Financial Services and Markets Act 2023 both include provisions establishing that DLT settlement can carry legal finality equivalent to that of traditional CSD settlement. Where such provisions do not exist, technical finality and legal finality remain misaligned, creating residual legal risk that institutions cannot price or manage efficiently.

Timing and synchronization. Legacy CSD settlement occurs at defined settlement cycles (T+1 in the US, T+2 across much of Europe). DLT platforms may settle atomically, intraday, or according to configurable schedules. A settlement abstraction layer must maintain a finality timestamp taxonomy capable of recording and comparing finality events across systems with fundamentally different temporal architectures.

Rail-agnosticism requires that finality norms be defined uniformly at the abstraction layer level, with each participating rail publishing its finality characteristics in a machine-readable format that the abstraction layer can evaluate. This is a governance requirement as much as a technical one.

Section 3 — Cross-Chain Equivalence

3.1 Definition and Distinction from Bridging

Core Concept — Definition

Cross-Chain Equivalence

Cross-chain equivalence is the principle that a digital representation of an asset on Chain A has identical economic, legal, and governance standing to its representation on Chain B, subject to a canonical proof attesting to their common origin and mutual exclusivity of claim. Equivalence is not a state that is established by technical means alone; it requires a legal recognition layer that treats the two representations as interchangeable for the purposes of ownership assertion, transfer, encumbrance, and enforcement.

The distinction between equivalence and bridging is fundamental and must be stated precisely. **Bridging** is a technical operation: it moves tokens — or representations of tokens — from one chain to another, typically by locking the original token on Chain A and minting a wrapped derivative on Chain B. The wrapped derivative is not the same instrument as the original; it is a claim against the bridge contract, not against the underlying asset. The legal standing of the bridged token is derivative of and dependent on the integrity of the bridge. Bridging moves tokens. **Equivalence asserts identity of the underlying claim.** Under an equivalence framework, the representation on Chain B is not a new instrument derived from the one on Chain A — it is the same instrument, re-instantiated on a different rail with an unbroken chain of provenance.

This distinction has profound practical consequences. A bridged token posted as margin collateral in a regulated derivatives transaction may not satisfy the eligibility requirements of the margin rule, because the regulator's eligibility schedule references the original instrument — a specific bond identified by ISIN — not a synthetic derivative of it. An equivalent token, carrying a canonical proof of identity with the original ISIN, satisfies the eligibility requirement by construction.

3.2 The Equivalence Certificate

The operational mechanism for asserting cross-chain equivalence is the **Equivalence Certificate**: a cryptographically anchored statement, issued and maintained by a trusted registry, attesting that two or more on-chain representations refer to the same underlying financial instrument and that no more than one of those representations can be asserted simultaneously as the basis for a rights claim.

An Equivalence Certificate contains at minimum: (a) the canonical identifier of the underlying instrument (ISIN, CUSIP, or equivalent, combined with a DLT namespace qualifier); (b) the chain identifiers and smart

contract addresses of all active representations; (c) an active representation flag specifying which representation currently carries the live rights claim; (d) a cryptographic hash of the most recent state of the underlying instrument's governance metadata; and (e) the identity and digital signature of the issuing registry, verified against a root of trust anchored in a regulated entity — typically the issuing CSD, the issuer's transfer agent, or a designated central registry.

Technical mechanisms supporting the issuance and verification of Equivalence Certificates include zero-knowledge proofs (ZKPs) for state equivalence — allowing a party to prove that a representation on Chain B is derived from the same state root as a representation on Chain A without revealing the full state; hash-linked canonical registries that maintain a content-addressed pointer to the current authoritative representation; and cross-chain messaging protocols (CCIP, the IBC protocol used in the Cosmos ecosystem) that can carry attestation payloads alongside token transfer messages.

3.3 Legal Frameworks and the Double-Claim Problem

The legal recognition of cross-chain equivalence is nascent but advancing. The ISDA Digital Asset Definitions (published 2023, updated 2025) provide a contractual framework for specifying the governing law and enforcement mechanism for digital asset transactions, including provisions for specifying the "controlling" representation in a multi-chain context. The EU's Markets in Crypto-Assets Regulation (MiCA), fully applicable from July 2026, establishes authorization and disclosure requirements for asset-referenced tokens and e-money tokens but does not yet provide a comprehensive framework for cross-chain equivalence recognition. US developments — including the GENIUS Act's federal stablecoin standards — address payment token issuance but leave the broader question of cross-chain asset equivalence to subsequent rulemaking.

The most acute legal challenge in cross-chain equivalence is the **double-claim problem**: the risk that two parties simultaneously assert ownership of the same underlying asset on the basis of representations held on different chains. In a bridge model, this risk manifests as a bridge exploit — if the locking contract on Chain A is compromised, the attacker can unlock the original token while the wrapped version on Chain B continues to circulate, creating two simultaneous claims on the same underlying value. In an equivalence model, the double-claim problem is addressed structurally: the Equivalence Certificate's active representation flag designates only one chain as the live rights claim holder at any moment. Changing the active representation requires a governance action by the issuing registry, which simultaneously deactivates the previous representation. Notary nodes — neutral, regulated entities that monitor the

state of all representations of a given instrument and attest to the uniqueness of the active claim — provide an additional layer of protection, complementing the technical mechanism with institutional oversight.

Section 4 — Canonical Digital Artifacts

4.1 Structure and Components

Core Concept — Definition

Canonical Digital Artifact (CDA)

A **Canonical Digital Artifact (CDA)** is the authoritative, uniquely identified on-chain or hybrid representation of a real-world financial instrument — whether a bond, a loan, an equity share, or a collateral claim — that serves as the primary evidentiary basis for the assertion of rights, the execution of lifecycle events, and the maintenance of audit continuity across all rails on which the instrument may be represented at any point in its lifecycle.

Canonicity is not merely a technical property; it is a governance designation. In a world where an instrument may simultaneously have representations on multiple rails, canonicity identifies which representation carries the authoritative state — the definitive record of ownership, encumbrance, and lifecycle history. The absence of a canonical representation in a multi-rail environment creates legal ambiguity (which record controls in a dispute?), operational risk (which system triggers the coupon payment?), and audit failure (which ledger does the regulator examine?).

A fully specified CDA consists of four structural components:

1. **Immutable Identifier:** A globally unique, stable identifier combining the instrument's conventional identifier (ISIN for bonds and equities, CUSIP for US securities) with a DLT namespace qualifier specifying the originating chain and smart contract address. This compound identifier remains constant across rail migrations and is resolvable by any compliant registry participant.

2. **Provenance Chain:** An ordered, append-only log of every material event in the instrument's lifecycle from issuance — primary issuance, each transfer of ownership, each encumbrance (pledge, repo, margin posting), each corporate action, and each rail migration — with each entry cryptographically linked to its predecessor. The provenance chain is the CDA's audit spine.
3. **Governance Metadata:** Machine-readable metadata specifying the controlling jurisdiction, the applicable legal framework for enforcement of rights, the current custodian and subcustodian chain, the smart contract address(es) of all active representations, and the identity of the Equivalence Certificate issuer. Governance metadata is updateable through a permissioned governance process, but each update is itself recorded in the provenance chain.
4. **Cryptographic Integrity:** A Merkle tree over the CDA's current state, with the root hash anchored on a high-assurance, widely witnessed ledger — a public blockchain, a regulated shared ledger, or a notary service. The Merkle structure allows any party to verify any component of the CDA's state without accessing the full artifact, and enables efficient proof of inclusion or exclusion.

4.2 Real-World Deployments

Several current deployments approximate the CDA model, though none yet implements the full specification described above. BlackRock's BUIDL tokenized money market fund — which has reached approximately \$2.9 billion in assets under management as of mid-2026 — operates across eight blockchain networks simultaneously, with Securitize serving as the designated transfer agent maintaining the canonical ownership record. BUIDL shares on each chain are representations of the same underlying fund share, not independent instruments; the canonical record is maintained off-chain by Securitize, with on-chain representations serving as the operational execution layer. This architecture approximates a CDA with an off-chain canonical registry.

Goldman Sachs' GS DAP (Digital Asset Platform) issues digital bonds as native digital instruments with Goldman Sachs as the operator of the canonical registry. The platform has been used for issuances by the European Investment Bank, the Hong Kong Monetary Authority, and others. DTCC's Project Whitney explored the use of a shared digital record for private market securities — an asset class where the absence of canonicity is particularly acute, given the reliance on paper-based cap tables and fragmented

transfer agent records. Each of these initiatives demonstrates demand for canonical representation; none yet provides a rail-portable CDA.

4.3 Lifecycle Management and Artifact Portability

The most operationally demanding aspect of CDA management is lifecycle event handling across rails. A bond CDA must correctly propagate a coupon payment — triggering cash delivery to the current holder, wherever that holder's record is maintained — without requiring the holder to be on the same rail as the payment agent. A CDA undergoing restructuring must update its governance metadata across all active representations atomically, or under a defined sequencing protocol that prevents divergent state between rails. Equity CDAs must handle corporate actions including dividends, rights offerings, stock splits, and mergers, each of which may affect the governance metadata, the provenance chain, or both.

Artifact portability — the CDA's ability to migrate from one rail to another without losing provenance, identity, or legal continuity — is the property that distinguishes a CDA from a rail-specific token. Portability is achieved through a migration protocol in which: (a) the owning party requests migration from the canonical registry; (b) the registry issues a migration certificate specifying the source rail, the destination rail, and the hash of the CDA's state at the moment of migration; (c) the CDA is deactivated on the source rail; (d) the CDA is instantiated on the destination rail, with the migration certificate appended to the provenance chain; and (e) the registry updates the Equivalence Certificate to reflect the new active representation. At no point does the migration break the provenance chain or alter the instrument's canonical identity.

Section 5 — Deterministic Replay

5.1 Definition and Regulatory Necessity

Core Concept — Definition

Deterministic Replay

Deterministic replay is the capacity of a settlement system to reconstruct, verify, and re-execute any historical settlement event using only the original inputs, producing bit-for-bit identical outputs on any

conformant execution environment, without dependence on system state that may have changed in the intervening period. Deterministic replay is the operational foundation of regulatory auditability in a multi-rail settlement environment.

The necessity of deterministic replay for regulated settlement is multi-dimensional. First, regulatory examination: the SEC, CFTC, FCA, ESMA, and their international counterparts require that regulated settlement systems maintain records sufficient to reconstruct any historical transaction. In a multi-rail environment, where a single economic transaction may span multiple systems, deterministic replay provides the mechanism by which any competent authority can verify, independently, that the transaction executed as recorded, on the rail identified, at the time claimed, with the outcome described. Second, dispute resolution: in cross-chain transactions, the facts in dispute may include the state of the blockchain at the moment of execution, the content of the smart contract invocation, and the finality status of the transaction at a specific timestamp. Deterministic replay provides a neutral, reproducible basis for resolving each of these questions. Third, system recovery and migration: when a rail is retired, upgraded, or replaced, deterministic replay allows historical transaction records to be migrated and verified on the successor system without loss of evidentiary integrity. Fourth, compliance with immutable record-keeping requirements including SEC Rule 17a-4 (which mandates that broker-dealer records be preserved in a non-rewriteable, non-erasable format for defined periods) and MiFID II transaction reporting obligations.

5.2 Technical Architecture and the Replay Audit Trail

The technical requirements for deterministic replay in a multi-rail environment are demanding but tractable. The foundational architectural pattern is **event sourcing**: rather than storing only the current state of the settlement system, every state change is recorded as an immutable, ordered event. The current state is always derivable by replaying the event log from the beginning — or from any verified checkpoint — which ensures that the system's history is self-contained and auditable.

Content-addressed storage — such as IPFS (InterPlanetary File System) or direct on-chain hash anchoring — ensures that the input data for any historical transaction is retrievable by its cryptographic hash, and that retrieval produces exactly the data that was present at the time of the original execution. Smart contract execution environments must be deterministic: the Ethereum Virtual Machine (EVM) is deterministic by design; Daml's virtual machine on the Canton Network provides deterministic execution

with privacy-preserving confidentiality; Hyperledger Fabric chaincode execution is deterministic within a defined endorsement context. Non-deterministic elements — particularly external oracle inputs, such as reference rates, FX prices, or eligibility data — must be frozen at the time of execution and recorded as part of the transaction's input set to preserve replay fidelity.

The **replay audit trail** is a structured log format that enables cross-rail verification. Each entry in the replay audit trail specifies: the event type (initiation, matching, pre-settlement, finality confirmation, rejection, exception); the timestamp in a canonical timezone-independent format (Unix epoch or ISO 8601 UTC); the initiating party's LEI; the rail identifier (a standardized string identifying the network, version, and governance context of the executing rail); the artifact hash (the Merkle root of the CDA's state at the time of transaction initiation); and the outcome hash (the Merkle root of the CDA's state following settlement, together with the hash of any cash instruments transferred). This structure allows any regulator or counterparty to verify the pre-state and post-state of every settlement event, on any rail, using only the replay audit trail and the content-addressed storage layer.

The principal challenges to deterministic replay in practice are three: time synchronization (clocks across different blockchain networks and legacy settlement systems may diverge by seconds or minutes, making precise timestamp comparisons unreliable without a canonical time oracle); non-deterministic oracle inputs (the solution is mandatory oracle input recording, as described above); and key management for historical signature verification (the signing keys used to authorize transactions may be rotated or revoked over time, requiring a key management archive that preserves the ability to verify historical signatures against the keys that were valid at the time of signing).

Regulatory Context

Eventual consistency models — common in distributed database architectures and some DLT implementations — are insufficient for regulated settlement. In an eventually consistent system, different nodes may observe different states of the ledger at a given moment, with the guarantee only that they will converge to the same state eventually. For regulatory audit, "eventually" is not a defined moment and cannot satisfy the precision requirements of examination. Regulated settlement demands strong consistency: every conformant node observes the same state at the same logical moment, enabling deterministic replay from any node in the network.

6.1 The Governance Gap

Core Concept — Definition

Multi-Rail Governance

Multi-rail governance is the set of legal agreements, technical standards, operational procedures, and supervisory frameworks that coordinate behavior across multiple settlement rails to ensure systemic consistency of outcomes, equitable access, prevention of concentration risk, and supervisory transparency — without requiring any single entity to control or own the coordination infrastructure.

The governance gap in multi-rail settlement is the most consequential impediment to the NVM framework's realization. Current interoperability efforts are, almost without exception, bilateral and ad hoc. The connection between SWIFT and the Canton Network, announced as part of SWIFT's Phase 3 tokenized asset transfer experiments, is a bilateral protocol between two specific parties. HQLAx's connection to Euroclear for collateral swaps is a bilateral arrangement governed by a bespoke legal agreement. There is no global, multilateral governance body with the authority, mandate, and technical capability to set binding standards for multi-rail settlement. The result is a web of bilateral agreements that is combinatorially explosive in complexity and that cannot scale to the number of rail pairings that institutional finance will require over the next decade.

6.2 The Multi-Rail Governance Stack

This paper proposes a **Multi-Rail Governance Stack** consisting of four distinct but interdependent layers, each addressing a different dimension of the governance gap. The layers are not independent: a failure at any layer propagates upward, undermining the coherence of the layers above it.

Layer	Name	Key Components	Responsible Parties
Layer 1	Standards Layer	ISO 20022 messaging; LEI for entity identification; ISIN + DLT namespace for asset identification; W3C DID for digital identity; common finality taxonomy	ISO, GLEIF, ANNA, W3C, SWIFT, BIS CPMI

Layer	Name	Key Components	Responsible Parties
Layer 2	Protocol Layer	Cross-chain communication protocols (CCIP, IBC); atomic settlement protocols; locked-asset notary services; Equivalence Certificate issuance infrastructure	Protocol operators (Chainlink, Cosmos), market infrastructure operators, regulated notary entities
Layer 3	Legal Layer	Governing law elections for cross-chain transactions; ISDA CSA amendments for DLT collateral; regulatory passporting for cross-border DLT settlement; EU DLT Pilot Regime extensions	ISDA, IOSCO, national securities regulators, national central banks, European Commission
Layer 4	Supervisory Layer	Central bank oversight extensions; tokenized deposit and CBDC settlement frameworks; mandatory reporting to supervisory data repositories; systemic risk monitoring and circuit-breaker protocols	BIS, Federal Reserve, ECB, Bank of England, FSB, national market supervisors

Layer 1 — Standards: The Standards Layer provides the common language through which all rails communicate. ISO 20022 is the natural backbone: its rich, structured messaging schema supports both legacy payment instruction formats and emerging digital asset transaction types. The Legal Entity Identifier (LEI), administered by the Global Legal Entity Identifier Foundation (GLEIF), provides a universal entity identification standard that is recognized by regulators in over 60 jurisdictions. The compound ISIN + DLT namespace identifier proposed in this paper extends the ISIN standard — maintained by the Association of National Numbering Agencies (ANNA) — to accommodate the multi-chain context. W3C Decentralized Identifiers (DIDs) provide a framework for self-sovereign digital identity that can complement LEI for natural person and pseudonymous party identification within DLT ecosystems.

Layer 2 — Protocol: The Protocol Layer provides the technical substrate for cross-rail value movement. Chainlink's CCIP and the IBC protocol (native to the Cosmos ecosystem) are the two most mature candidates for the cross-chain messaging role. Neither is currently designed with the regulatory requirements of institutional settlement in mind — CCIP, for example, lacks native support for ISO 20022 message encapsulation — but both provide the foundational messaging and token transfer capabilities that a settlement protocol layer requires. The Protocol Layer also encompasses atomic settlement protocols — mechanisms ensuring that the cash leg and asset leg of a transaction either both settle or neither settles, eliminating principal risk — and locked-asset notary services that maintain the active representation state for Equivalence Certificates.

Layer 3 — Legal: The Legal Layer addresses the most complex and politically sensitive dimension of multi-rail governance: the question of which law governs a cross-chain transaction, and how that law is enforced. The ISDA Credit Support Annex (CSA) — the standard documentation for OTC derivatives collateral — requires amendment to recognize DLT-native collateral as eligible margin, and to address the governing law questions raised by cross-chain collateral posting. Regulatory passporting — the recognition by one jurisdiction of another jurisdiction's regulatory authorization for settlement infrastructure — is essential for cross-border multi-rail transactions, and is currently absent in any systematic form outside of bilateral mutual recognition agreements.

Layer 4 — Supervisory: The Supervisory Layer extends existing central bank and regulatory oversight frameworks to the multi-rail environment. BIS Project Agora — a collaboration between seven central banks and a large number of private financial institutions — is exploring the integration of tokenized commercial bank deposits with tokenized central bank money in a unified settlement environment, representing the most ambitious current attempt to build Supervisory Layer infrastructure. BIS Project Mariana tested automated market makers for cross-border CBDC settlement between the Swiss, Singapore, and French central banks, demonstrating that supervisory frameworks can be designed to encompass novel settlement mechanisms without abandoning the prudential objectives of the existing regime.

6.3 Central Banks as Neutral Settlement Anchors

In any multi-rail governance architecture, the question of what serves as the neutral settlement asset — the medium of final discharge that is acceptable to all counterparties and all rails — is decisive. The natural answer is tokenized central bank money. Central bank liabilities are uniquely positioned as the neutral settlement asset in a multi-rail world: they carry no credit risk, are accepted by all regulated financial institutions, and carry the sovereign authority of the issuing central bank. Tokenized central bank money — whether in the form of a wholesale CBDC, a tokenized central bank deposit, or a tokenized reserve balance — provides the common settlement medium that allows cash legs of cross-rail transactions to achieve finality on a basis that all participants recognize as definitive.

The deployment of tokenized central bank money at scale remains a work in progress. The Federal Reserve's FedNow system, while not itself a CBDC, provides a foundation for 24/7 wholesale dollar settlement that could be extended to support tokenized deposit settlement. The European Central Bank has conducted wholesale CBDC experiments as part of the Eurosystem's exploratory work. The Bank of

England is developing its Digital Pound framework. Multi-rail governance must prevent the concentration risk that would arise if a single orchestration entity — whether a private platform or a central infrastructure operator — became a single point of failure for the entire multi-rail ecosystem. Circuit-breaker protocols, mandatory redundancy in orchestration infrastructure, and limits on the market share of individual orchestration providers are essential components of a prudentially sound supervisory framework.

Section 7 — Neutral Value Movement as an Institutional Framework

7.1 The NVM Operational Layers

The five concepts developed in Sections 2 through 6 — rail-agnostic settlement, cross-chain equivalence, canonical digital artifacts, deterministic replay, and multi-rail governance — are not independent innovations. They are mutually reinforcing components of a single, integrated framework: Neutral Value Movement. Their relationship can be organized into three operational layers, each dependent on the layer below it and enabling the layer above it.

NVM Layer	Core Components	Function
Layer 1: Representation	Canonical Digital Artifacts + Cross-Chain Equivalence	Establishes the authoritative identity, provenance, and cross-rail standing of financial instruments. Ensures that value has a portable, legally coherent representation independent of any specific rail.
Layer 2: Execution	Rail-Agnostic Settlement + Deterministic Replay	Provides the operational mechanisms for moving value across rails with outcome equivalence, and for recording those movements in a format that supports regulatory audit, dispute resolution, and system recovery.
Layer 3: Governance	Multi-Rail Governance Stack (Standards, Protocol, Legal, Supervisory)	Coordinates the behavior of all participants across all rails through binding standards, legal frameworks, and supervisory oversight — ensuring systemic consistency and preventing the emergence of new fragmentation at the governance level.

The layered structure reflects a design principle: higher layers cannot function without the integrity of lower layers, but lower layers are insufficient without the higher layers that give them institutional meaning. Canonical Digital Artifacts without cross-chain equivalence are merely well-structured tokens. Rail-agnostic settlement without deterministic replay is operationally capable but regulatorily opaque.

The Multi-Rail Governance Stack without the Representation and Execution Layers has nothing to govern. All three layers must be implemented simultaneously, or the framework degrades into something indistinguishable from today's bilateral, ad hoc interoperability efforts.

This simultaneous requirement is the most important distinction between NVM and prior interoperability proposals. Interoperability projects — including those cited in this paper — typically address one layer in isolation: a messaging protocol here, a legal amendment there, a pilot experiment connecting two specific platforms. NVM requires a coherent architectural commitment across all three layers from the outset. This is a higher bar, but it is the only bar that produces rail-agnosticism as a durable property of the settlement system rather than as an incidental feature of a specific bilateral connection.

7.2 Institutional Readiness and Transition Risks

The institutional readiness path for NVM compliance varies by entity type but shares a common sequence. A **Tier 1 bank** must first establish an internal canonical registry capable of issuing and maintaining CDAs for all instruments it holds or transfers on behalf of clients — a function that today resides partially in custody systems, partially in trading systems, and partially in the client's own records, with no unified canonical view. It must then extend its legal documentation to encompass cross-chain equivalence — amending ISDA master agreements, CSAs, and custody agreements to recognize DLT-native representations — and connect to at least one compliant abstraction-layer protocol. A **central securities depository** must develop the capability to issue Equivalence Certificates for instruments in its custody, maintain the active representation registry, and provide the replay audit trail infrastructure for all transactions processed through its systems. A **central bank** must deploy tokenized central bank money as a settlement asset accessible to multi-rail participants, extend its supervision to encompass the orchestration entities and protocol operators that constitute the Protocol Layer, and contribute to the development of the Supervisory Layer through the BIS and FSB frameworks.

Transition risks are substantial and must be addressed directly. **Legacy system entanglement** is the most immediate: the settlement systems of major CSDs and custodians are deeply integrated with risk management, accounting, regulatory reporting, and client-facing systems built over decades, and cannot be replaced wholesale. The NVM transition requires a coexistence period — potentially five to ten years — during which legacy systems and NVM-compliant infrastructure operate in parallel, with CDAs serving as the bridge between them. **Regulatory jurisdictional conflicts** arise when a cross-chain transaction spans instruments governed by different national laws, each of which may have different requirements

for finality recognition, record-keeping, or collateral eligibility. The Legal Layer of the Multi-Rail Governance Stack addresses this systematically, but until the Legal Layer is established, jurisdictional conflicts will be resolved case by case. **Network effect lock-in** by first-mover platforms — particularly JPMorgan Kinexys, Broadridge DLR, and SWIFT — represents a concentration risk: if these platforms capture the majority of institutional transaction flow before NVM standards are established, they may have both the incentive and the market power to resist standardization that would reduce their competitive advantage.

The evidence that NVM-scale infrastructure is achievable is now convincing. JPMorgan Kinexys processing \$1.5 trillion in cumulative transactions, Broadridge DLR handling \$385 billion in daily repo volume, and BlackRock BUIDL achieving \$2.9 billion in AUM across eight blockchains are not proof-of-concept demonstrations: they are production deployments at institutional scale. The question is no longer whether tokenized settlement infrastructure can work; it is whether it will be built with the rail-agnostic architecture that allows these platforms to interoperate, or whether each platform will become a new silo — more efficient than the legacy CSD, but no less fragmented.

Section 8 — Policy Recommendations

The following five recommendations are addressed collectively to regulators, central banks, market infrastructure operators, and standards bodies. They are sequenced in order of dependency: each recommendation enables those that follow.

Recommendation 1

Mandate Canonical Asset Identifiers Across All DLT Settlement Platforms

Rationale: The absence of a universal, machine-readable asset identifier that spans both the conventional ISIN namespace and the DLT namespace is the single most tractable impediment to the Representation Layer of the NVM framework. Without it, no Equivalence Certificate can be issued, no CDA can be uniquely identified across rails, and no settlement abstraction layer can match positions held on different platforms. **Responsible Parties:** ANNA (Association of National Numbering Agencies), ISO TC68, SWIFT, national regulators with jurisdiction over DLT settlement platforms, and market

infrastructure operators. **Implementation Timeline:** Pilot standard by Q4 2026; mandatory adoption for regulated platforms by end 2028. **Expected Outcome:** A compound ISIN + DLT namespace identifier, compatible with ISO 6166 (ISIN), ISO 20022 (messaging), and the W3C DID specification, that uniquely identifies any financial instrument representation on any compliant rail and is resolvable by any authorized participant.

Recommendation 2

Establish a Legal Framework for Cross-Chain Equivalence Recognition via ISDA / IOSCO Joint Working Group

Rationale: Cross-chain equivalence cannot be achieved by technical means alone. The legal recognition that a DLT-native representation of an ISIN-identified instrument carries the same rights, obligations, and enforceability as its CSD-held counterpart requires explicit legislative or regulatory action in each material jurisdiction. A joint ISDA / IOSCO working group — with participation from the CFTC, SEC, ESMA, FCA, and MAS — is the appropriate forum for developing the model legal provisions, documentation standards, and regulatory recognition mechanisms that the Legal Layer requires. **Responsible Parties:** ISDA, IOSCO, national securities regulators, European Commission (MiCA delegated acts). **Implementation Timeline:** Working group constituted by Q1 2027; model provisions published by Q4 2027; national adoption by 2029. **Expected Outcome:** A globally consistent legal framework under which a regulated Equivalence Certificate constitutes sufficient proof of asset identity for the purposes of margin eligibility, beneficial ownership assertion, and enforcement of rights across all adhering jurisdictions.

Recommendation 3

Require Deterministic Replay Capability for All Regulated Settlement Systems

Rationale: As regulated settlement migrates to DLT platforms and multi-rail architectures, the existing record-keeping frameworks — SEC Rule 17a-4, MiFID II transaction reporting, EMIR trade reporting — were designed for single-system, centralized infrastructure and do not address the replay requirements

of distributed execution. Regulators should amend record-keeping rules to require, for all regulated settlement systems, the maintenance of a complete, content-addressed replay audit trail conforming to the standard defined in Section 5.2 of this paper. **Responsible Parties:** SEC, CFTC, ESMA, FCA, national supervisors. **Implementation Timeline:** Proposed rulemaking by Q2 2027; final rules by Q4 2028; compliance deadline Q4 2030 (to allow legacy systems a realistic migration window). **Expected Outcome:** Every regulated settlement event, on any rail, is reconstructable by any competent authority using the replay audit trail, eliminating the evidentiary gap that currently exists for cross-chain transactions.

Recommendation 4

Create a Multi-Rail Supervisory Council Under BIS Auspices

Rationale: The Multi-Rail Governance Stack's Supervisory Layer requires a multilateral coordinating body with the authority to set binding supervisory standards for multi-rail settlement infrastructure, monitor systemic risk across the ecosystem, and coordinate regulatory responses to emerging threats. The BIS — through its Committee on Payments and Market Infrastructures (CPMI) and its Financial Stability Board (FSB) coordination role — is the natural host for such a body. The Multi-Rail Supervisory Council would not replace national supervisors but would coordinate their oversight of entities and protocols that operate across jurisdictions, analogous to the role of the Basel Committee in banking supervision.

Responsible Parties: BIS, G20 central banks, FSB member authorities. **Implementation Timeline:** Founding charter adopted by G20 Finance Ministers by mid-2027; Council operational by end 2027.

Expected Outcome: A permanent, adequately resourced multilateral body with a defined mandate, observer rights over multi-rail infrastructure operators, and the authority to develop binding supervisory standards — filling the governance gap that is the single largest systemic risk in the current multi-rail ecosystem.

Recommendation 5

Accelerate Tokenized Central Bank Money Deployment as the Neutral Settlement Asset

Rationale: The NVM framework's Execution Layer requires a neutral settlement asset that is accepted by all rails and carries no credit risk. Tokenized central bank money is the only instrument that satisfies these requirements. Central banks should accelerate the deployment of wholesale tokenized central bank money — whether through CBDC programs, tokenized reserve programs, or enhanced access to central bank accounts for regulated DLT platforms — to ensure that the neutral settlement asset is available before multi-rail settlement volume reaches systemic scale. The BIS Project Agora framework, connecting seven central banks with private financial institutions, provides the most mature existing template. **Responsible Parties:** Federal Reserve, ECB, Bank of England, Bank of Japan, People's Bank of China, Swiss National Bank, BIS Innovation Hub. **Implementation Timeline:** Wholesale tokenized central bank money pilots operational in at least three major currency areas by end 2027; production deployment by 2030. **Expected Outcome:** A globally accessible, credit-risk-free settlement asset in major currencies, available to all NVM-compliant settlement rails, eliminating the reliance on commercial bank credit as the de facto settlement medium in tokenized markets and materially reducing systemic credit risk in multi-rail settlement.

Section 9 — Conclusion

The settlement modernization imperative of the 2020s is not primarily a technological challenge. The technology — distributed ledger platforms, atomic settlement protocols, zero-knowledge proofs, cross-chain messaging — has advanced to a state of institutional readiness that is no longer seriously in question. JPMorgan Kinexys, Broadridge DLR, BlackRock BUIDL, HQLAx, and the SWIFT tokenized asset transfer experiments collectively demonstrate that tokenized settlement infrastructure can operate at institutional scale, with institutional reliability, under institutional governance. The challenge is architectural: ensuring that the platforms and networks being built today are designed from first principles to be rail-agnostic rather than rail-specific, and that the legal and supervisory frameworks that govern them are coherent across rails rather than fragmented by them.

This paper has argued that Neutral Value Movement provides the correct architectural foundation for that challenge. Its three operational layers — the Representation Layer (Canonical Digital Artifacts and

Cross-Chain Equivalence), the Execution Layer (Rail-Agnostic Settlement and Deterministic Replay), and the Governance Layer (the Multi-Rail Governance Stack) — are individually tractable and collectively sufficient. Each layer addresses a distinct dimension of the rail-dependency problem: representation ensures that value has a portable identity; execution ensures that value can move across rails with outcome equivalence and regulatory auditability; governance ensures that the entire ecosystem operates under a coherent, multi-jurisdictional framework that prevents the emergence of new fragmentation at the coordination layer.

The 2026–2030 settlement infrastructure horizon will be decisive. The institutional deployments underway today will accumulate network effects, legal precedents, and regulatory accommodations that will shape the architecture of global settlement for decades. If those deployments are built with NVM principles embedded from the outset — if canonicity, equivalence, rail-agnosticism, deterministic replay, and multi-rail governance are treated as first principles rather than future enhancements — the outcome will be a settlement system of unprecedented efficiency, resilience, and regulatory transparency. If they are not, the tokenized settlement landscape of 2030 will be a more sophisticated version of today's fragmentation: faster rails, smarter contracts, and equally siloed value. The choice, and the window in which it can be made, belongs to this moment.

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